

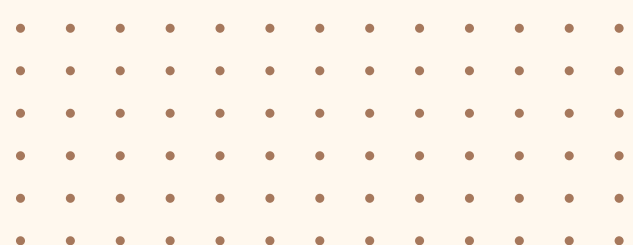


CLASS 10 PHYSICS – LIGHT

(BOARD LEVEL QUESTIONS)

SECTION A: VERY SHORT ANSWER (1 MARK)

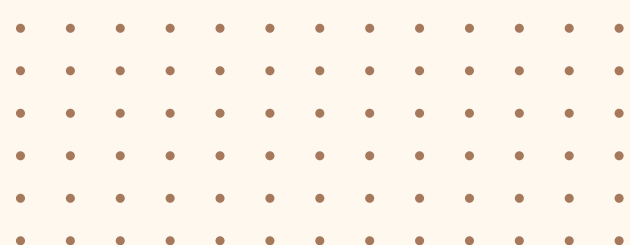
- 1. Define the principal focus of a concave mirror.**
- 2. What is meant by radius of curvature of a spherical mirror?**
- 3. Define magnification produced by a mirror.**
- 4. What is a real image?**
- 5. What is a virtual image?**
- 6. Define refractive index of a medium.**
- 7. State Snell's law of refraction.**
- 8. What is meant by power of a lens?**
- 9. Write the SI unit of power of a lens.**
- 10. Define optical centre of a lens.**





SECTION B: SHORT ANSWER (2–3 MARKS)

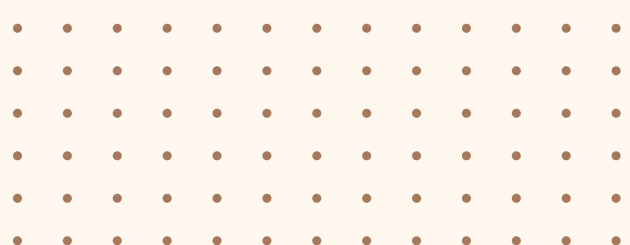
- 1. State the laws of reflection of light.**
- 2. Write any four characteristics of the image formed by a plane mirror.**
- 3. Why is a convex mirror preferred as a rear-view mirror in vehicles?**
- 4. Define focal length of a spherical mirror.**
- 5. Explain the phenomenon of refraction of light.**
- 6. What happens to a ray of light when it passes from a rarer to a denser medium?**
- 7. Define refractive index of a medium in terms of speed of light.**
- 8. Write the mathematical expression for refractive index.**
- 9. Define power of a lens and write its formula.**
- 10. What is magnification? Write its formula for spherical mirrors.**





SECTION C: RAY DIAGRAM QUESTIONS (3–4 MARKS)

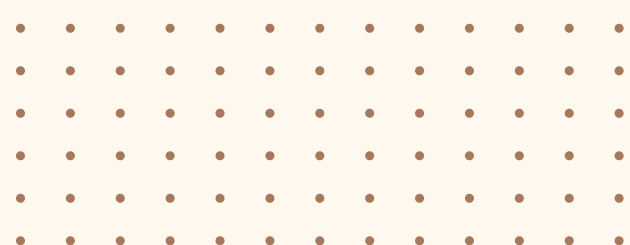
- 1. Draw a ray diagram to show image formation by a concave mirror when the object is at infinity.**
- 2. Draw a ray diagram for image formation when the object is placed beyond the centre of curvature of a concave mirror.**
- 3. Draw a ray diagram when the object is placed between the focus and centre of curvature of a concave mirror.**
- 4. Draw a ray diagram when the object is placed between the pole and focus of a concave mirror.**
- 5. Draw a ray diagram to show image formation by a convex mirror.**
- 6. Draw a ray diagram for a convex lens when the object is placed beyond $2F$.**
- 7. Draw a ray diagram for a convex lens when the object is between F and $2F$.**
- 8. Draw a ray diagram for a concave lens showing image formation.**
- 9. Draw a labelled diagram to show refraction through a rectangular glass slab.**
- 10. Show lateral displacement in a glass slab using a ray diagram.**





SECTION D: NUMERICAL PROBLEMS (3–5 MARKS)

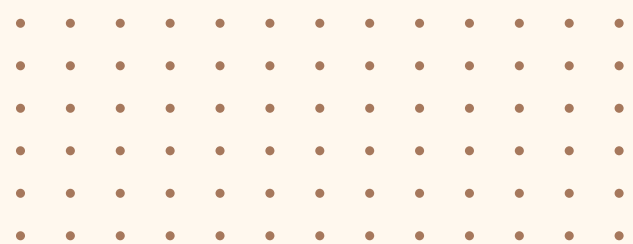
1. An object is placed at a distance of 20 cm in front of a concave mirror of focal length 10 cm. Calculate the position of the image.
2. Calculate the magnification for the above case.
3. A convex mirror has a focal length of 15 cm. An object is placed at a distance of 30 cm. Find the position of the image.
4. A convex lens has a focal length of 20 cm. An object is placed at a distance of 40 cm. Find the position of the image.
5. Calculate the power of a lens whose focal length is 50 cm.
6. A lens has a power of +2 dioptre. Calculate its focal length.
7. Find the magnification when the image distance is 30 cm and object distance is –10 cm.
8. An image is formed at a distance of 25 cm by a concave mirror. Calculate the magnification.
9. A lens forms an image twice the size of the object. Find the magnification.
10. Calculate the refractive index of a medium if the angle of incidence is 30° and the angle of refraction is 20° .





SECTION E: LONG ANSWER (5 MARKS)

- 1. Derive the mirror formula for a spherical mirror.**
- 2. Derive the lens formula for a thin lens.**
- 3. Explain the New Cartesian sign convention with the help of a diagram.**
- 4. Describe the structure and working of the human eye.**
- 5. Explain the common defects of vision and their correction.**
- 6. Explain myopia with the help of a ray diagram. How is it corrected?**
- 7. Explain hypermetropia with the help of a ray diagram. How is it corrected?**
- 8. What is presbyopia? How can it be corrected?**
- 9. Explain refraction of light through a rectangular glass slab with a diagram.**
- 10. Define power of a lens. Explain its significance and uses.**





ASSERTION-REASON

Directions:

For each of the following questions, choose the correct option:

- (a) Both A and R are true and R is the correct explanation of A
- (b) Both A and R are true but R is not the correct explanation of A
- (c) A is true but R is false
- (d) A is false but R is true

Q1.

Assertion (A): A concave mirror can form a real image.

Reason (R): Light rays actually meet after reflection.

Q2.

Assertion (A): A convex mirror always forms a virtual image.

Reason (R): Reflected rays diverge after reflection.

Q3.

Assertion (A): Myopia is corrected using a concave lens.

Reason (R): A concave lens diverges light rays.

Q4.

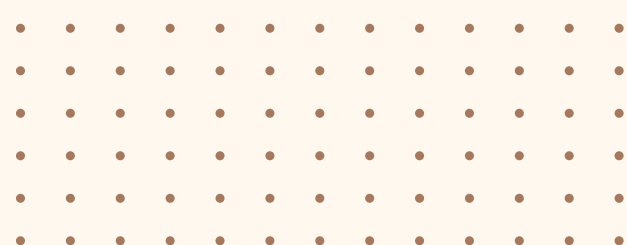
Assertion (A): The power of a convex lens is positive.

Reason (R): The focal length of a convex lens is positive.

Q5.

Assertion (A): Refraction of light occurs due to change in speed of light.

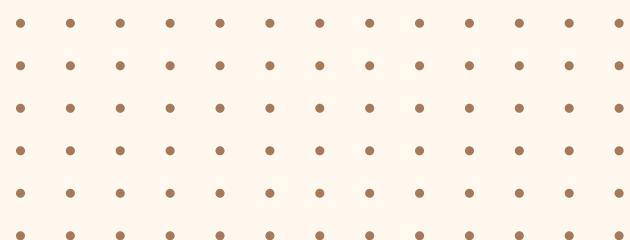
Reason (R): The speed of light is the same in all media.





EXAM TIP (BOARD LEVEL)

Attempt ray diagrams neatly with pencil
Always follow sign convention in numericals
Write formula + substitution + final answer with unit
In theory answers → write points, not paragraphs



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INSPIRES